

Customer Table:

- **customer_id** (Primary Key): Unique identifier for each customer.
- **customer_name**: Name of the customer.
- **customer_address**: Address of the customer.
- **customer_email**: Email address of the customer.
- **customer_phone**: Phone number of the customer.
- **customer_type**: Type of customer (e.g., individual, business).
- **customer_membership_level**: Membership level of the customer (if applicable).

Retailer Table:

- **retailer_id** (Primary Key): Unique identifier for each retailer.
- **retailer_name**: Name of the retailer.
- **retailer_address**: Address of the retailer.
- **retailer_email**: Email address of the retailer.
- **retailer_phone**: Phone number of the retailer.
- **retailer_type**: Type of retailer (e.g., grocery store, convenience store).
- **retailer_inventory**: Inventory details or products sold by the retailer.

Supplier Table:

- **supplier_id** (Primary Key): Unique identifier for each supplier.
- **supplier_name**: Name of the supplier.
- **supplier_address**: Address of the supplier.
- **supplier_email**: Email address of the supplier.

- **supplier_phone:** Phone number of the supplier.
- **supplier_type:** Type of supplier (e.g., manufacturer, distributor).
- **supplier_products:** Products supplied by the supplier.

Transaction Table:

- **transaction_id** (Primary Key): Unique identifier for each transaction.
- **customer_id** (Foreign Key): Reference to the Customer table.
- **retailer_id** (Foreign Key): Reference to the Retailer table.
- **product_id** (Foreign Key): Reference to the Product table.
- **quantity:** Quantity of the product purchased.
- **transaction_date:** Date of the transaction.

Location Table:

- **location_id** (Primary Key): Unique identifier for each location.
- **location_name:** Name of the location (e.g., Alabama, California).
- **address:** Address of the location.

Product_Location Table (Many-to-Many Relationship):

- **product_id** (Foreign Key): Reference to the product table.
- **location_id** (Foreign Key): Reference to the location table.
- **manufacturing_date:** Date when the product was manufactured at the location.

- **expiry_date:** Expiry date of the product manufactured at the location.
- **batch_number:** Batch number or lot number of the product manufactured at the location.

Distribution Center Table:

- **distribution_center_id** (Primary Key): Unique identifier for each distribution center.
- **distribution_center_name:** Name of the distribution center.
- **distribution_center_address:** Address of the distribution center.
- **distribution_center_manager:** Name or identifier of the distribution center manager.
- **distribution_center_capacity:** Capacity or storage size of the distribution center.
- **distribution_center_contact:** Contact information for the distribution center (phone, email).
- **distribution_center_inventory:** Details of the inventory or products stored at the distribution center.
- **distribution_center_operations_hours:** Operating hours of the distribution center.
- **distribution_center_transportation:** Transportation methods used for distribution (e.g., trucks, vans).
- **distribution_center_security:** Security measures in place at the distribution center.

Product Table:

- **product_id** (Primary Key): Unique identifier for each product.

- **product_name:** Name of the product.
- **product_type:** Type or category of the product (e.g., soft drinks, noodles).
- **product_description:** Description or details about the product.
- **product_manufacturer:** Reference to the Manufacturer table indicating the manufacturer of the product.
- **product_supplier:** Reference to the Supplier table indicating the primary supplier of the product.
- **product_price:** Price of the product.
- **product_quantity:** Quantity or stock level of the product.
- **product_expiry_date:** Expiry date of the product.
- **product_location:** Location where the product is manufactured or stored.
- **product_status:** Status of the product (e.g., available, out of stock).
- **product_image:** Image or photo of the product.

Supplier Orders Table:

- **supplier_order_id** (Primary Key): Unique identifier for each supplier order.
- **supplier_id:** Reference to the Supplier table indicating the supplier who placed the order.
- **manufacturer_id:** Reference to the Manufacturer table indicating the manufacturer receiving the order.
- **order_date:** Date and time when the order was placed.
- **order_status:** Status of the order (e.g., pending, confirmed, shipped).
- **order_total:** Total amount of the order.

Manufacturer Orders Table:

- **manufacturer_order_id** (Primary Key): Unique identifier for each manufacturer order.
- **manufacturer_id**: Reference to the Manufacturer table indicating the manufacturer who placed the order.
- **supplier_id**: Reference to the Supplier table indicating the supplier fulfilling the order.
- **order_date**: Date and time when the order was placed.
- **order_status**: Status of the order (e.g., pending, confirmed, shipped).
- **order_total**: Total amount of the order.

Customer Orders Table:

- **order_id** (Primary Key): Unique identifier for each order.
- **customer_id**: Reference to the Customer table indicating the customer who placed the order.
- **order_date**: Date and time when the order was placed.
- **order_status**: Status of the order (e.g., pending, shipped, delivered).
- **order_total**: Total amount of the order.

1. Supplier Distribution Orders Table:

- **supplier_distribution_order_id** (Primary Key): Unique identifier for each supplier distribution order.
- **order_id**: Reference to the Order table indicating the original order.
- **from_manufacturer_id**: Identifier of the manufacturer from which the order is being distributed.
- **to_supplier_id**: Identifier of the supplier to which the order is being distributed.
- **distribution_date**: Date and time when the distribution order was executed.
- **status**: Status of the distribution order (e.g., pending, in transit, delivered).

Orders Table:

- **retailer_distribution_order_id** (Primary Key): Unique identifier for each retailer distribution order.
- **order_id**: Reference to the Order table indicating the original order.
- **from_supplier_id**: Identifier of the supplier from which the order is being distributed.
- **to_retailer_id**: Identifier of the retailer to which the order is being distributed.
- **distribution_date**: Date and time when the distribution order was executed.
- **status**: Status of the distribution order (e.g., pending, in transit, delivered).

Customer Distribution Orders Table:

- **customer_distribution_order_id** (Primary Key): Unique identifier for each customer distribution order.
- **order_id**: Reference to the Order table indicating the original order.
- **from_retailer_id**: Identifier of the retailer from which the order is being distributed.
- **to_customer_id**: Identifier of the customer to whom the order is being distributed.
- **distribution_date**: Date and time when the distribution order was executed.
- **status**: Status of the distribution order (e.g., pending, in transit, delivered).

1. Customer Table:

- One customer can place many orders (One-to-Many relationship with Order table).

2. **Retailer Table:**

- One retailer can have many distribution orders (One-to-Many relationship with Retailer Distribution Orders table).

3. **Supplier Table:**

- One supplier can have many distribution orders (One-to-Many relationship with Supplier Distribution Orders table).

4. **Transaction Table:**

- One transaction is associated with one customer, one retailer, and one product (Many-to-One relationship with Customer, Retailer, and Product tables).

5. **Location Table:**

- Many products can be stored or manufactured at many locations (Many-to-Many relationship with Product table through Product_Location table).

6. **Product Table:**

- One product can have many distribution orders (One-to-Many relationship with Retailer Distribution Orders, Supplier Distribution Orders, and Customer Distribution Orders tables).
- One product can be associated with one manufacturer (Many-to-One relationship with Manufacturer table).
- One product can have one primary supplier (Many-to-One relationship with Supplier table).

7. **Distribution Center Table:**

- Many products can be stored or distributed from one distribution center (One-to-Many relationship with Product table).

8. **Manufacturer Orders Table:**

- One manufacturer can place many orders to many suppliers (Many-to-Many relationship with Supplier table through Manufacturer Orders table).

9. Supplier Orders Table:

- One supplier can place many orders to many manufacturers (Many-to-Many relationship with Manufacturer table through Supplier Orders table).

10. Customer Orders Table:

- One customer can place many orders (One-to-Many relationship with Order table).

11. Supplier Distribution Orders Table:

- One supplier can have many distribution orders (One-to-Many relationship with Supplier table).
- One distribution order can be sent from one manufacturer to one supplier (Many-to-One relationship with Manufacturer table).

12. Retailer Distribution Orders Table:

- One retailer can have many distribution orders (One-to-Many relationship with Retailer table).
- One distribution order can be sent from one supplier to one retailer (Many-to-One relationship with Supplier table).

13. Customer Distribution Orders Table:

- One customer can have many distribution orders (One-to-Many relationship with Customer table).
- One distribution order can be sent from one retailer to one customer (Many-to-One relationship with Retailer table).

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